

BrainLift: Teaching the LSAT as a Skill, Not a Fact Base

An LSAT study app built on the idea that the modern LSAT rewards transferable reasoning schemas, not memorized content.

Owners

- Arthur Xu
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Purpose

The purpose of this BrainLift is to redefine how an LSAT study tool should be built. One of the key observations here is that the modern LSAT is roughly two-thirds Logical Reasoning, and every Logical Reasoning problem is unique. There is almost nothing to *memorize*. What has to improve is a transferable reasoning skill. This is ability to recognize, under time pressure, which underlying pattern a never-before-seen argument uses, and to apply the right procedure to it.

Thus, the flashcard aspect of this learning app will not be the traditional memorization technique. It will instead focus on rewarding *transfer*. This is the ability of a user to perform well on novel problems that share some underlying structure with other problems, but differ on nearly every surface feature. The central goal of this BrainLift will be to assemble a proper learning science that focuses on a few things: schema abstraction, discrimination, transfer, and speed-accuracy tradeoff.

In Scope

- Learning science governing transfer, schema induction, discrimination learning, retrieval practice, scaffolding/fading, and the speed–accuracy tradeoff, as they apply specifically to a reasoning exam.
- A structured categorization of LSAT patterns. This includes question types, flaw types, trap (wrong-answer) types, and reading-comprehension structures. These are the key ideas to master.
- Developing defensible spiky positions on how an LSAT tool should differ from a conventional flashcard system.

- Light references to how these principles would surface in an app (a schema-weighted study queue, separate measurement of memory vs. performance, a falsifiable feature test).

Out of Scope

- Logic Games / Analytical Reasoning. These were permanently removed from the LSAT as of August 2024 and replaced by a second Logical Reasoning section. There is no point to focus on this point.
- Memory-model research as a centerpiece (FSRS, spacing schedules). The LSAT has too little memorizable content for a memory model to be the core; that work is relevant only to a small portion of the learning app and is treated as supporting rather than foundational.
- Full engineering specification of the app. This document is learning-science context.

A note about the LSAT: The majority of the LSAT will be Logical Reasoning in today's testing environment. This means around $\frac{2}{3}$ of the test will be on Logical Reasoning. This means each problem will be unique. They will have their own distinct problem to read through and questions to answer. However, there will be underlying patterns and concepts present despite these problems being distinct. The model, then, will rely on identifying these different concepts and honing the student's skills within each of these concepts.

DOK 4: Spiky Points of View (SPOVs)

Spiky POV 1: The LSAT is a transfer problem, so a flashcard app cannot be used in a traditional memory retention manner.

Explanation: The flashcard structure and the memory-scheduling algorithm built upon it assumes the thing being learned is a stable item that will reappear and must be retained. That assumption holds for the MCAT's fact base. It fails for the LSAT, where the item (like a specific argument about coffee, or voting, or paleontology) will literally never be seen again. So what is recurring and what the user needs to understand is the deep structure. This is a correlation-causation flaw, a sufficient/necessary confusion, or an out-of-scope trap. Building the app around memory retention and memorization will make students excellent at recalling explanations they have already seen, but will not help them extrapolate on new problems. The correct unit of mastery is the **schema**: the pattern that transfers. Every other position in this BrainLift follows from taking this seriously. The practical consequence is that the app's data model, its study queue, and its scoring must all be keyed to schema mastery, not card retention. Thus, the metric key to the app is its ability to *measure the gap* between "can recall this card" and "can answer a new question with this structure."

Quick Ideas:

- Students are much better suited to answering new questions when they can abstract the concepts from the problem. This is best achieved by solving multiple problems with the same concepts/abstractions. Students will better be able to glean the patterns in this case rather than doing so from a single problem.
https://pdf.retrievalpractice.org/transfer/Gick_Holyoak_1980.pdf
- Comparison forces them to extract the underlying schema.
- The app will thus have many concepts and patterns and have students identify which patterns and concepts they contain. Flashcards will be used for these types of purposes.

Spiky POV 2: Mastery should occur and be measured at the level of the *schema*, which starts with the flaws of a problem rather than question type.

Explanation: Prep companies organize the LSAT primarily by *question type* (Assumption, Strengthen, Weaken, Inference...), because question type is what the stem announces and is therefore easy to label. But question type is a surface feature of the *task*, not of the *reasoning*. Again, the reasoning is the most important part. Flaws are thus an important, more transferable unit because identifying flaws means having a deep understand of the problem. Just like teaching is the highest form of learning, being able to articulate confidently and precisely why other solutions are incorrect is a key part of a student solving a problem. Indeed, a handful of flaws (correlation/causation, sufficient/necessary, part/whole, sampling error, equivocation) recur *across* question types. Therefore, a tool that diagnoses weakness and helps student develop better reasoning helps the student more. This is a spiky position because it inverts the standard prep-company information architecture: I am claiming the *secondary* tag (flaw) should be the primary organizing axis of study and measurement. Then the question-type tag comes next. A deep understanding of wrong answers will help students quickly identify the correct answer on tests.

Quick Ideas:

- Building upon the previous idea, trap-identification is another important factor in correctly arriving at a correct answer. Again, each problem will have wrong answers that follow similar traps that students need to avoid. Flashcards are great for students to identify these trap. This will also help them identify why a solution is wrong and develop deeper understandings of the answers.

Spiky POV 3: For LSAT explanation-reading, scaffolding should fade *dramatically more slowly* than the standard consensus prescribes. For the two-answer decision, it should very nearly never fade.

Explanation: The well-established consensus from worked-example research (Sweller) is that instructional scaffolding should fade as a learner gains expertise, because support that helps a novice becomes redundant, even harmful, for an expert. I accept the *direction* of this finding but argue it is miscalibrated for the LSAT. I believe that on hard LSAT items, the difficulty does not disappear with expertise. It relocates to the final two-answer decision. A shared consensus

among practice books and elite test-takers is that problems often come down to two decisions, where one attractive trap answer and one correct answer remain. Understanding the distinction between them not only precisely but quickly is one of the things that will separate a 175-scorer as for a 160-scorer. The expertise-reversal effect assumes the expert no longer benefits from seeing the reasoning, but on the LSAT the expert is still routinely losing points at precisely the step the explanation addresses. Therefore the explanation of *why the runner-up answer is wrong and the winner is right* retains its value far longer than fading theory predicts. Thus, fading should occur slowly and not disappear at all for these binary decisions. The falsifiable version will keep raising accuracy on hard (two-answer) items at a point where fading theory predicts it should plateau.

Spiky POV 4: Speed is not a separate skill to train after accuracy; accuracy measured without latency is a vanity metric on a timed reasoning test.

Explanation: Most prep advice treats timing as a final-phase concern: get accurate first, then get fast. I agree with this sentiment in the beginning portions of student training. However, there must come a point when the student gets uncomfortable with the timing. The key goal is to *change* the students' reasoning process. If they are given infinite time to think and don't work under time pressure, they will not narrow down the items they must consider on the actual test. Again, this narrowing requires expertise and deep understanding of the problem (SPOV 2). On a test with brutal per-section timing, an answer that is correct but too slow is, in aggregate, a *wrong* answer, because it consumes the budget that would have earned points elsewhere. A readiness model that reports accuracy without jointly modeling response latency is reporting a number the real exam will not honor. A student who is 95% accurate at 2x the time budget is *less ready* than one who is 85% accurate within budget, and an honest tool must say so and flag the imbalance rather than flattering the accurate-but-slow student with a high score.

Experts:

- **Robert A. Bjork & Elizabeth L. Bjork (and Nate Kornell)**
 - **Who:** Cognitive psychologists at UCLA. They are the originators of the "desirable difficulties" framework and are prominent figures in research on interleaving and spacing.
 - **Focus:** Why conditions that make learning feel *harder* and slower (interleaving, spacing, retrieval) produce better long-term and transferable learning than blocked, fluent practice. Kornell & Bjork (2008) showed interleaving improved learners' ability to *categorize* novel exemplars the discrimination skill at the heart of LR.
<https://web.williams.edu/Psychology/Faculty/Kornell/Publications/Kornell.Bjork.2008a.pdf>

- **Why Follow:** The LSAT's core demand is "which pattern is this?". This a discrimination task. The Bjork line of work is the direct scientific basis for interleaving question/flash types rather than drilling them in blocks.
- **Where:** Bjork Learning and Forgetting Lab, UCLA (bjorklab.psych.ucla.edu); Kornell & Bjork (2008), "Learning concepts and categories: Is spacing the enemy of induction?", *Psychological Science*.
- **Mary L. Gick & Keith J. Holyoak**
 - **Who:** Cognitive scientists; foundational researchers on analogical transfer and schema induction.
 - **Focus:** Their 1980/1983 work showed that people fail to transfer a solution principle to a structurally identical new problem when they have seen only one example, but succeed when they compare *two* analogs sharing deep structure. Thus, comparison forced extraction of the underlying schema.
 - **Why Follow:** This is the scientific core of SPOV 1 and 2 and of the "contrasting cases" study method. I need students to stop solving the specific argument and start recognizing the reusable pattern.
 - **Where:** Holyoak's UCLA Reasoning Lab; Gick & Holyoak (1980), "Analogical Problem Solving", *Cognitive Psychology* 12; Gick & Holyoak (1983), "Schema Induction and Analogical Transfer", *Cognitive Psychology* 15.
 - https://pdf.retrievalpractice.org/transfer/Gick_Holyoak_1980.pdf
- **Jeffrey D. Karpicke & Henry L. Roediger III**
 - **Who:** Cognitive psychologists; leading figures in research on the testing/retrieval-practice effect.
 - **Focus:** Karpicke & Roediger (2008), "The Critical Importance of Retrieval for Learning" (*Science* 319, 966–968), demonstrated that repeated *retrieval* drives durable learning while repeated *study* after initial success does little.
 - Also, learners' confidence is uncorrelated with actual retention. Related work (Karpicke & Blunt, 2011) shows retrieval practice beats elaborative study even on transfer-style assessments.
 - **Why Follow:** This will reframe retrieval for the LSAT. The thing to be retrieved is not a fact but the *execution of a reasoning procedure*. This is will my AI features will come in. Also the scientific basis for the app measuring true performance rather than trusting student confidence. It is often the case that the students misjudge their own readiness.
 - **Where:** Karpicke's lab, Purdue (learninglab.psych.purdue.edu); Karpicke & Roediger (2008), *Science*, DOI 10.1126/science.1152408.
 - <http://www.sciencemag.org/about/permissions.dtl>
- **John Sweller**
 - **Who:** Educational psychologist; originator of Cognitive Load Theory and the worked-example effect.
 - **Focus:** Novices learn complex problem-solving better from studying worked examples than from unguided problem-solving. This scaffolding should fade as expertise grows (the expertise-reversal effect).

- **Why Follow:** Sweller's consensus is the position SPOV 3 acknowledges but does not fully accept. Understanding the worked-example and fading literature precisely is what makes the contrarian claim defensible rather than ignorant. I argue that the LSAT's difficulty structure should allow for scaffolding throughout the process.
 - **Where:** Widely published; foundational papers in *Cognitive Science* (1988) and *Educational Psychology Review*; see also Kirschner, Sweller & Clark (2006).
 - <https://doi.org/10.1016/B978-0-12-387691-1.00002-8>
 - **Doug Rohrer**
 - **Who:** Cognitive psychologist (University of South Florida) focused on interleaving in mathematics and the durability of practice schedules.
 - **Focus:** Interleaved practice of problem *types* improves students' ability to select the correct solution method on later mixed tests. This will be difficult but important for students to master early.
 - **Why Follow:** Rohrer's category-discrimination findings map almost one-to-one onto the LR section.
 - **Where:** uweb.cas.usf.edu/~drohrer; Rohrer & Taylor (2007) and Rohrer (2012) on interleaving.
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DOK 3: Insights

Theme A: The unit of mastery

- **Insight 1:** Because every Logical Reasoning item is surface-unique, the only thing that can be "mastered" and reused is the deep structure. This means the app's atomic object should be a *schema* (a flaw, a trap, a question-type procedure), and individual questions are merely *instances* used to train and test that schema. This is a different data model from every flashcard app, where the card is the atom.
- **Insight 2:** Flaw structures are the highest-leverage schemas because they are *shared across* question types. The important categories here are Flaw, Weaken, Necessary Assumption, and Parallel-Flaw performance. So organizing study and diagnosis by flaw gives the most transfer per unit of study. This is similar to the consensus that teaching is one of the best ways of learning as it forces one to articulate their reasonings and fully explain wrong answers. (Synthesizes Gick & Holyoak's schema-transfer with the LSAT taxonomy.)

Theme B: How schemas are actually acquired

- **Insight 3:** A single worked example does not produce transfer; *comparison of two analogs* does (Gick & Holyoak). Therefore the app should deliberately present items in

contrasting pairs/sets. These are two arguments with the same flaw but different topics, or one with a flaw and one cleverly without. Then the prompt is to have students articulate what they share.

- **Insight 4:** Interleaving beats blocking specifically when the skill includes *identifying which category an item belongs to* (Kornell & Bjork; Rohrer). Since LR's first move is "what type is this?", interleaving question/flaw types is matched to the cognitive demand of the problem.
- **Insight 5:** Retrieval, not restudy, is what consolidates (Karpicke & Roediger). For the LSAT the retrieved object is a *procedure*, so effective practice is making the student *execute* the reasoning from memory, then checking it. This is also where AI comes in. AI has not generally been good at evaluating reasoning and commentary. It can point out specific errors and tell students where they went wrong, but does not point out patterns of weaknesses. Training AI on strong training data and allowing it to point out *patterns* in student reasoning is more important.

Theme C: Scaffolding and the two-answer fork

- **Insight 6:** On hard LSAT items, students often experience that there are two answers to make a decision between.. This is the key empirical observation behind SPOV 3, and it explains why even high scorers keep losing points at the same step. Fading theory implicitly assumes the expert no longer benefits from the explanation, but the expert is still missing exactly the thing the explanation addresses.
- **Insight 7:** Therefore scaffolding should be *decomposed*, not faded uniformly.

Theme D: Traps, weighting, and speed

- **Insight 8:** Wrong answers are not random. Rather, there are a small set of recurring **trap types** (out-of-scope, too-strong, reversed relationship, half-right, premise-restatement, etc.). This means "why is this wrong?" is itself a learnable and transferrable schema. Therefore, modeling *which trap a student habitually falls for* is a more diagnostic signal than *which question they missed*. (Novel application of schema-transfer logic to the distractor side of problems.)
- **Insight 9:** Choosing the right answer is partly a *weighting* problem. In other words, deciding whether vagueness, scope, or strength is the decisive feature on this item. Re-encountering items that reward the *same weighting* builds the pattern recognition that lets a student apply that weighting to a novel item. This connects schema induction to the answer-selection step specifically, not just to flaw-spotting.

- **Insight 10:** On a brutally timed test, latency and accuracy are not separable measures of readiness; a correct-but-slow answer trades away points elsewhere. An honest readiness model must treat response time as co-equal with correctness, and must flag the accurate-but-slow student rather than rewarding them with a high score. (Basis for SPOV 4.)
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DOK 2: Knowledge Tree

The structured foundation: domain broken from broad categories down to specific facts and summaries, with sources.

- **Category 1: The modern LSAT — structure and what it measures**
 - **Subcategory 1.1: Current format (post-Aug 2024)**
 - **Source:** LSAC, "About the LSAT" / "Types of LSAT Questions"
 - **DOK 1 — Facts:**
 - Three scored sections: two Logical Reasoning (24–26 questions each) and one Reading Comprehension (26–28 questions), each 35 minutes.
 - Logic Games (Analytical Reasoning) permanently removed August 2024, replaced by a second LR section.
 - Scored on 120–180 scale; raw score (number correct) is equated for form difficulty, not curved against other test-takers; no guessing penalty.
 - An unscored experimental section (LR or RC) is indistinguishable from scored sections during the test.
 - **DOK 2 — Summary:**
 - The test is ~2/3 Logical Reasoning, so LR drives most of the score and is the right primary target.
 - The exam measures reasoning *skill* applied to novel material, not recall of a body of content — which is the whole premise of this BrainLift.
 - **Link to source:**
<https://www.lsac.org/lSAT/prepare/types-lsat-questions>
- **Category 2: The schemas to be mastered (LSAT pattern taxonomy)**
 - **Subcategory 2.1: LR question types**
 - **Source:** Prep-taxonomy synthesis (PowerScore, Manhattan, The LSAT Trainer, Cambridge LSAT)
 - **DOK 1 — Facts:**

- ~10–14 types: Identify the Flaw; Necessary Assumption; Sufficient Assumption; Strengthen; Weaken; Evaluate; Must Be True/Inference; Most Strongly Supported; Main Conclusion; Method of Reasoning; Parallel Reasoning; Parallel Flaw; Point at Issue/Agreement; Paradox; Principle (Identify/Apply); Role of a Statement; plus "EXCEPT" variants.
 - Assumption + Flaw + Inference $\approx$ 40% of LR; adding Strengthen, Weaken, Paradox, Principle $\approx$ 75%+.
 - **DOK 2 — Summary:**
 - Question type is the surface/task axis; useful for labeling but not the deepest organizing unit (see SPOV 2).
 - **Link to source:**
<https://www.cambridgelsat.com/resources/information/question-types/>
- **Subcategory 2.2: Flaw types (the transferable backbone)**
 - **Source:** Khan Academy LSAT "Types of flaws"; LSATHacks
 - **DOK 1 — Facts:**
 - Causal: correlation→causation, reversed causation, common-cause overlooked, post hoc.
 - Conditional: necessary/sufficient confusion, mistaken reversal, mistaken negation.
 - Sampling/evidence: unrepresentative/biased/small sample, appeal to ignorance.
 - Quantity/scope: part↔whole, percentage vs. absolute number, equivocation.
 - Structure: circular reasoning, ad hominem, straw man, false dilemma, appeal to authority/emotion.
 - **DOK 2 — Summary:**
 - The same flaw recurs across many question types, making flaws the highest-transfer schema and the right primary diagnostic axis.
 - **Link to source:**
<https://lsathacks.com/logical-reasoning-question-types/>
- **Subcategory 2.3: Trap (wrong-answer) types**
 - **Source:** Prep-taxonomy synthesis
 - **DOK 1 — Facts:**
 - Out-of-scope; too-strong/extreme; too-weak; opposite; reversed relationship; half-right/half-wrong; premise restatement; could-be-true (fails must-be-true bar); real-world-plausible-but-unsupported; right-answer-to-wrong-question.
 - RC-specific: distortion of author's view, wrong-viewpoint attribution, scope too broad/narrow, tone mismatch.

- **DOK 2 — Summary:**
 - Distractors are systematic, so "why wrong" is a learnable schema; tracking a student's habitual trap is highly diagnostic (Insight 8).
 - **Link to source:**
<https://lsathacks.com/logical-reasoning-question-types/>
- **Category 3: Learning-science foundations**
 - **Subcategory 3.1: Schema induction / transfer**
 - **Source:** Gick & Holyoak (1980, 1983)
 - **DOK 1 — Facts:** Single example rarely transfers to a structurally identical new problem; comparing two analogs sharing deep structure produces transfer by forcing schema extraction.
 - **DOK 2 — Summary:** Foundation for the "contrasting cases" method and the schema-as-unit data model.
 - **Link to source:**
https://pdf.retrievalpractice.org/transfer/Gick_Holyoak_1980.pdf
 - **Subcategory 3.2: Interleaving / discrimination**
 - **Source:** Kornell & Bjork (2008); Rohrer & Taylor (2007)
 - **DOK 1 — Facts:** Interleaving exemplars improved categorization of novel items vs. blocking, despite feeling harder and lowering practice-phase performance; interleaving problem types improves later method-selection.
 - **DOK 2 — Summary:** Matched to LR's "which type is this?" demand; the practice-phase difficulty is the desirable difficulty.
 - **Link to source:**
<https://web.williams.edu/Psychology/Faculty/Kornell/Publications/Kornell.Bjork.2008a.pdf>
 - **Subcategory 3.3: Retrieval practice**
 - **Source:** Karpicke & Roediger (2008), *Science* 319, 966–968
 - **DOK 1 — Facts:** Repeated testing produced a large positive effect on delayed recall; repeated studying after initial success did not; students' confidence was uncorrelated with actual performance.
 - **DOK 2 — Summary:** For the LSAT, the retrieved object is a procedure; also justifies measuring real performance over student confidence.
 - **Link to source:** <https://doi.org/10.1126/science.1152408>
 - **Subcategory 3.4: Worked examples, fading, expertise reversal**
 - **Source:** Sweller (Cognitive Load Theory); Kirschner, Sweller & Clark (2006)
 - **DOK 1 — Facts:** Novices benefit from studying worked examples; support that aids novices can hinder experts (expertise-reversal); consensus is to fade scaffolding as expertise grows.

- **DOK 2 — Summary:** The consensus SPOV 3 pushes against; the LSAT's difficulty relocates to the two-answer fork rather than disappearing, so contrastive scaffolding should fade far more slowly.
- **Link to source:** Kirschner, Sweller & Clark (2006), *Educational Psychologist* 41(2).

Sources I want to review:

<https://dl.acm.org/doi/epdf/10.1145/3534678.3539081>

<https://github.com/open-spaced-repetition/awesome-fsrs/wiki/Spaced-Repetition-Algorithm:-A-Three%E2%80%90Day-Journey-from-Novice-to-Expert>